INSTITUTE OF PATHOLOGY & GENOMIC MEDICINE

CLINICAL SURGICAL PATHOLOGY & BIOMARKER REPORT

Patient Name:	Sophia Rodriguez	**Record ID (MRN):**	P1003
Date of Birth:	1974-03-22	**Biological Sex:**	Female
Ordering Physician:	Dr. Oncologist	**Facility NPI:**	1234567890
Diagnosis:	Malignant neoplasm of sigmoid colon	**Report Date:**	2026-05-18

1. Surgical Pathology Diagnostic Summary

Colonoscopy identified sigmoid mass; staging CT showed multiple hepatic and peritoneal implants.

2. Immunohistochemistry & Molecular Biomarker Profile

Biomarker / Gene	Result / Status	Testing Method	Specimen / Clinical Finding
KRAS_STATUS	Wild Type / Negative	IHC / NGS	KRAS codons 12, 13, 61, 146: No mutation detected.
NRAS_STATUS	Wild Type / Negative	IHC / NGS	NRAS codons 12, 13, 61: No mutation detected.
BRAF_STATUS	Negative	IHC / NGS	BRAF V600E: Negative.
MSI_MMR_STATUS	MSI-High / dMMR	IHC / NGS	IHC shows loss of MLH1 and PMS2 nuclear expression. MSI-High confirmed by PCR.
TUMOR_LOCATION	Left-sided (Sigmoid)	IHC / NGS	Primary tumor located in sigmoid colon (left side).

3. Pathologist Attestation & Molecular Review

This diagnostic report has been verified by the attending surgical pathologist. Results are calibrated against CLIA/CAP accredited laboratory standards and integrated into the master oncology electronic record.

Certified By: James Wilson, MD, FCAP (Chief of Molecular Pathology) • Sign-off Date: 2026-05-18